Course notes

Definition and properties

The exponential is continuous and strictly increasing on $\mathbb{R}$, and its range is $]0,+\infty[$. For every $x>0$, there is a unique real number y such that $e^y = x$: we denote it by $\ln x$.

$$\ln(e^y) = y \quad (y \in \mathbb{R}), \qquad e^{\ln x} = x \quad (x>0), \qquad \ln 1 = 0, \quad \ln e = 1.$$

For $a, b > 0$ and $n \in \mathbb{Z}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a, \quad \ln\sqrt{a} = \tfrac{1}{2}\ln a.$$

Proof of the product rule. $e^{\ln a + \ln b} = e^{\ln a} e^{\ln b} = ab$; the uniqueness of the preimage gives the formula. There is no analogous formula for $\ln(a+b)$.

Derivative, variations and limits

We accept that ln is differentiable on $]0,+\infty[$. Differentiating $e^{\ln x} = x$ gives $e^{\ln x}(\ln x)' = 1$; hence $(\ln x)' = \frac{1}{x} > 0$. ln is strictly increasing and concave, because its second derivative is $-\frac{1}{x^2}$.

$$\lim_{x \to 0^+} \ln x = -\infty, \qquad \lim_{x \to +\infty} \ln x = +\infty.$$

Its curve is the reflection of that of the exponential in the line $y = x$. The line $x = 0$ is a vertical asymptote. Its tangent at 1 gives $\ln x \leq x - 1$ for $x > 0$.

Composition and comparison of growth rates

If $u > 0$ is differentiable, $(\ln u)' = \frac{u'}{u}$. For every integer $m \geq 1$,

$$\lim_{x \to +\infty} \frac{\ln x}{x^m} = 0, \qquad \lim_{x \to 0^+} x^m \ln x = 0.$$

Proof at 0. Set $t = -\ln x$; as $x \to 0^+$, $t \to +\infty$ and $x \ln x = -te^{-t} \to 0$ by comparison of growth rates. For $m \geq 1$, the same reasoning gives $-te^{-mt} \to 0$. At $+\infty$, set $t = \ln x$ to write $\frac{\ln x}{x^m} = \frac{t}{e^{mt}} \to 0$.

Equations and thresholds

For $A, B > 0$, $\ln A = \ln B \Leftrightarrow A = B$ and $\ln A \leq \ln B \Leftrightarrow A \leq B$. Always start with the domain.

Example. $\ln(x-1)+\ln(x+1)=\ln 8$ requires $x>1$. The equation becomes $x^2-1=8$, so $x=3$ only.

Threshold. $0.8^n \leq 0.1$ is equivalent to $n\ln 0.8 \leq \ln 0.1$, then $n \geq \frac{\ln 0.1}{\ln 0.8} \simeq 10.319$. The first integer is 11.

Example of a study. For $f(x)=\frac{\ln x}{x}$, $f'=\frac{1-\ln x}{x^2}$; f increases on $]0,e]$, decreases on $[e,+\infty[$, and its maximum is $\frac{1}{e}$. Its limit at 0^+ is $-\infty$, its limit at $+\infty$ is 0.

